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CSA6501 - Generative AI

Lab programs (20 – 25)

21. Product Description (Smart Fitness Watch)

Program

```
product = input("Product: ")
features = input("Features: ")
print("\nZero-shot:")
print(f"Write a product description for {product} with features: {features}.")
print("\nOne-shot:")
print("Example: Wireless Earbuds -> Compact earbuds with Bluetooth and long battery life.")
print(f"Now write for {product}.")
print("\nFew-shot:")
print("Example1: Laptop -> Fast and powerful.\nExample2: Smartphone -> Stylish and feature-rich.")
print(f"Now write for {product}.")
```

Output



```
from transformers import pipeline

# Load the pre-trained sentiment analysis pipeline
classifier = pipeline("sentiment-analysis")

# Get input from the user
text = input("Enter a sentence: ")

# Perform sentiment analysis
result = classifier(text)

# Display the result
print("\nSentiment Analysis Result:")
print("Label: ", result[0]["label"])
print("Score: ", round(result[0]["score"], 4))
```

[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 713406.
Using a pipeline without specifying a model name and revision in production is not recommended.
loading weights: 0% [0/1000000] 0s/eta [unknown], 100%
Enter a sentence: I am very happy with the service.

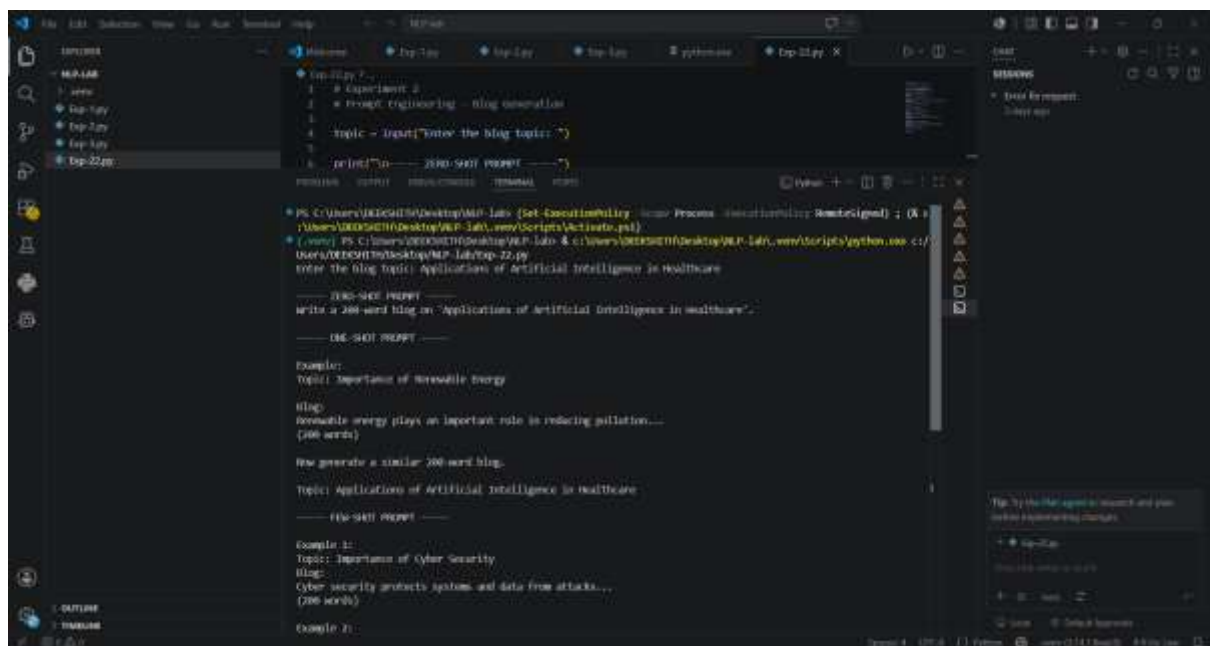
Sentiment analysis result:
label: POSITIVE
score: 0.8899

22. Design Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog on "Applications of Artificial Intelligence in Healthcare."

Program

```
topic = input("Enter the blog topic: ")
print("\n----- ZERO-SHOT PROMPT -----")
print(f"Write a 200-word blog on '{topic}'.")
print("\n----- ONE-SHOT PROMPT -----")
print(f"Topic: {topic}")
print("\n----- FEW-SHOT PROMPT -----")
print(topic)
```

Output



23. Design Zero-shot, One-shot, and Few-shot prompts to summarize an article into 50 words.

Program

```
article = input("Paste the article:\n")

print("\n----- ZERO-SHOT PROMPT -----")

print("Summarize the following article in exactly 50 words.")

print(article)

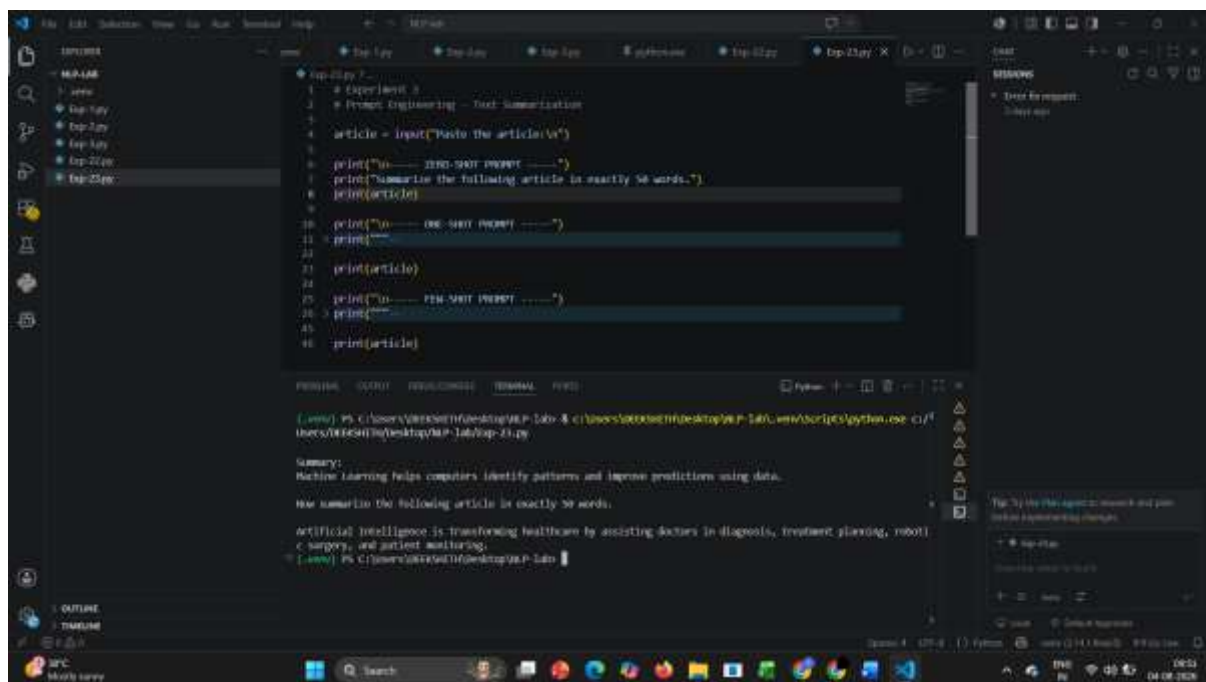
print("\n----- ONE-SHOT PROMPT -----")

print(article)

print("\n----- FEW-SHOT PROMPT -----")

print(article)
```

Output



The screenshot shows a Python IDE with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code editor contains the following Python code:

```
1 # Experiment 3
2 # Prompt Engineering - Text Summarization
3
4 article = input("Paste the article:\n")
5
6 print("\n----- ZERO-SHOT PROMPT -----")
7 print("Summarize the following article in exactly 50 words.")
8 print(article)
9
10 print("\n----- ONE-SHOT PROMPT -----")
11 print(article)
12
13 print(article)
14
15 print("\n----- FEW-SHOT PROMPT -----")
16 print(article)
17
18 print(article)
```

The terminal at the bottom shows the output of the program. It displays the prompt "Paste the article:\n" followed by the user input "Machine learning helps computers identify patterns and improve predictions using data. Now summarize the following article in exactly 50 words. Artificial intelligence is transforming healthcare by assisting doctors in diagnosis, treatment planning, robotic surgery, and patient monitoring." The terminal also shows the file path and the command to run the script.

24. Design Zero-shot, One-shot, and Few-shot prompts to generate a professional leave email.

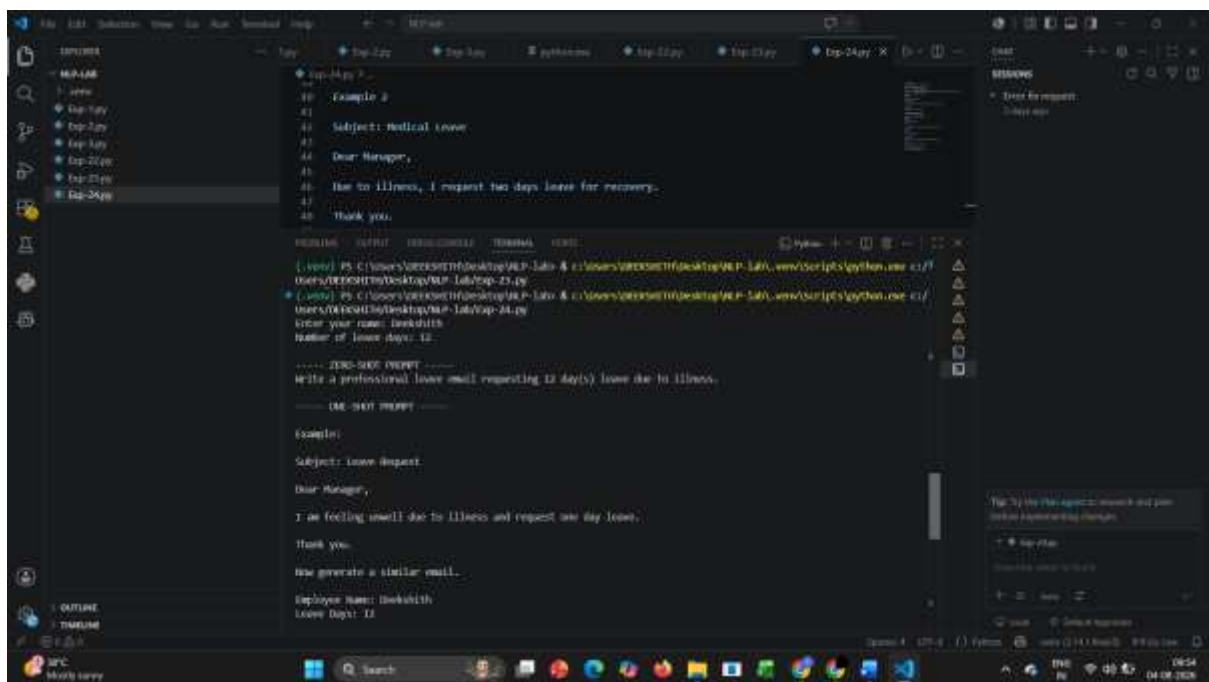
Program

```
name = input("Enter your name: ")
days = input("Number of leave days: ")
print("\n----- ZERO-SHOT PROMPT -----")
print(f"Write a professional leave email requesting {days} day(s) leave due to illness.")

print("\n----- ONE-SHOT PROMPT -----")
print(f"Employee Name: {name}")
print(f"Leave Days: {days}")

print("\n----- FEW-SHOT PROMPT -----")
print(f"Employee Name: {name}")
print(f"Leave Days: {days}")
```

Output



25. Design Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop.

Program

```
event = input("Enter workshop name: ")

print("\n----- ZERO-SHOT PROMPT -----")

print(f"Write a promotional social media post for '{event}'.")

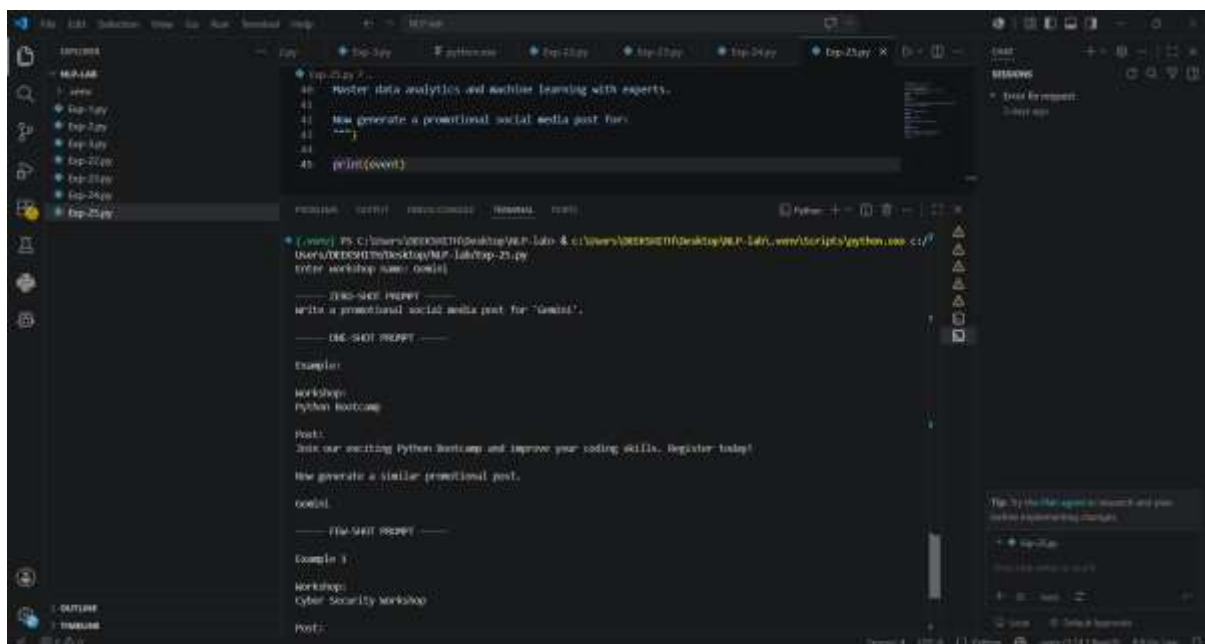
print("\n----- ONE-SHOT PROMPT -----")

print(event)

print("\n----- FEW-SHOT PROMPT -----")

print(event)
```

output



The screenshot shows a Python IDE with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code editor contains a Python script with comments and a function definition. The terminal shows the output of the script, including the prompts for zero-shot, one-shot, and few-shot prompts, and the generated social media post for the workshop name 'Genial'.

```
40: """Prompt: data analytics and machine learning with experts.
41:
42: Now generate a promotional social media post for:
43: """
44:
45: print(event)
```

PS C:\Users\DEEPMITH\Desktop\MLP-Lab & c:\Users\DEEPMITH\Desktop\MLP-Lab\venv\Scripts\python.exe c:/Users/DEEPMITH/Desktop/MLP-Lab/top-25.py
Enter workshop name: Genial

----- ZERO-SHOT PROMPT -----
Write a promotional social media post for 'Genial'.

----- ONE-SHOT PROMPT -----

Example:
Workshop:
Python Bootcamp

Post:
Join our exciting Python Bootcamp and improve your coding skills. Register today!

Now generate a similar promotional post.

Genial.

----- FEW-SHOT PROMPT -----

Example 1:
Workshop:
Cyber Security Workshop

Post: